Peter Shamoun
petershamoun.com | linkedin.com/in/peter-shamoun | petershamoun80@gmail.com

Education

University of California, San Diego — San Diego, CA
Bachelor of Science in Data Science — *March 2026*
Coursework: Data Structures & Algorithms, Machine Learning, NLP, Database Management, Cloud Computing

Experience

AI Engineer — June 2026 – Present
Xifin — *San Diego, CA*
- Developing agentic AI workflows to prevent insurance claim denials, and autonomously appeal them.

Data Science Intern — June 2025 – Feb 2026
LPL Financial — *San Diego, CA*
- Developed NLP-based keyword extraction and sentiment analysis pipeline on advisor feedback from business findings reports, identifying organizational trends and actionable insights across the company.
- Built production-deployed bulk closure detection system using Pandas/NumPy to identify suspicious alert resolution patterns through statistical analysis, preventing revenue leakage and enabling compliance audits.

Undergraduate AI Researcher — May 2025 – Present
HDSI, Data Smith Lab — *San Diego, CA*
- Leading the research of a novel AI safety technique using task vector algebra to strategically fine-tune large language models, precisely manipulating model behavior to mitigate harmful outputs from jailbreak attempts.

Data Science Intern — Sep. 2024 – May 2025
Data Science Alliance — *San Diego, CA*
- Developed geospatial optimization models in Python to identify optimal food bank placements, creating automated data pipelines with the US Census/ACS API, and brought data driven recommendations to the city of San Diego.
- Built and deployed an interactive web dashboard with ArcGIS API to visualize model results, enabling stakeholders to perform real-time scenario analysis via interactive sliders.

Data Science Intern — Jun. 2024 - Sep. 2024
Nexon America — *Los Angeles, CA*
- Engineered a data pipeline in Snowflake to process large-scale, longitudinal player data; trained and deployed clustering models using AWS SageMaker to predict Lifetime Value and inform player engagement strategy.
- Developed a sentiment analysis tool by fine-tuning a foundation model accessed via AWS Bedrock; implemented a RAG pipeline to summarize unstructured community feedback, delivering actionable insights to product teams.

Research Assistant — Sep. 2023 - Jan. 2024
San Diego Supercomputer Center — *San Diego, CA*
- Developed and optimized a TensorFlow-based sequential neural network to compress high-dimensional plasma physics data, reducing matrix storage requirements and improving computational efficiency.

Projects

Cheat Sheet AI | *Python, NLP, Flask, Docker, Oracle Cloud* — Jan. 2025
- Developed an AI-powered study tool using fine-tuned LLMs and prompt engineering to generate personalized cheat sheets, serving over **2000+** users
- Designed and deployed a scalable, containerized Flask backend using Docker on Oracle Cloud to ensure high availability and serve production traffic.

Graph Theory Visualizer | *React.js, D3.js, HTML/CSS* — Dec. 2024
- Developed an interactive D3.js web app to visualize graph algorithms (BFS, DFS, Dijkstra's); officially adopted as an instructional tool in UCSD's DSC40B course.

Technical Skills

Languages: Python, SQL, Java, JavaScript, R, MATLAB
Libraries/Frameworks: TensorFlow, Keras, Scikit-learn, Pandas, NumPy, Flask, React, D3.js, Matplotlib
Tools & Cloud: Docker, LangChain, AWS (SageMaker, Bedrock), Oracle Cloud, Snowflake, Tableau, Git/GitHub